RESEARCH BUILD / V0.2.0

One recording contract. Two separate models.

Measure a reference-defined tongue hold during practice. Detect reference-labelled respiratory-event candidates in recorded audio.

INPUT BOUNDARY	Aligned physical-unit samples + explicit validity + independent labels + acquisition provenance
SHARED STORAGE	Checksummed array chunks and session manifest Raw arrays remain available. Labels never enter feature vectors.
SEPARATE ANALYSIS	Awake: EMG features → random forest Night: energy / MFCC statistics → logistic regression
OUTPUT	Valid probability or explicit abstention → candidate timeline → held-out research report

Implemented boundary

Local saved-session analysis, baseline training, JSON model export, replay and participant-isolated evaluation. CPU execution; no cloud service required by this design.

Evidence boundary

No clinical recordings were trained in this build. Synthetic tests establish software behavior only. Neither model establishes treatment benefit or readiness for patient feedback.

> **Architectural commitment:** overnight predictions cannot select exercises or change treatment. The models share storage, never clinical conclusions or labels.

A recording must explain what every sample means.

Session time is an explicit common clock. Phone arrival time is not a synchronization method.

CONTRACT	REQUIRED EVIDENCE
Session identity	Participant, session and placement IDs; source and version; permission statement; explicit synthetic flag; acquisition configuration and label source.
Stream identity	Sample rate, ordered channel names, per-channel units, start time, sample count, chunk positions and SHA256 hashes. Optional calibration travels with the stream.
Signal chain	Training requires nonempty per-stream configuration. Rate, channels, units and signal-chain contents form the frozen input signature.
Reference labels	Nonempty interval, task, positive / negative / exclude state and provenance. Scored intervals cannot overlap within a task. Exclusions may override them.
Clock evidence	Training requires a documented reference-alignment method and declared maximum error no greater than half the output bin. At the default 100 ms bin, that engineering guard is 50 ms.

Storage and failure behavior

Numeric `.npy` chunks preserve supplied values and data types. Boolean arrays carry per-sample validity. The reader verifies checksums and shapes, rejects object arrays and escaping paths, and reads bounded windows through a small mapping cache.

Physical units must be converted correctly upstream. Preserve original ADC counts as a separate stream when needed. Missing or out-of-range samples remain invalid; the reader does not invent data or resample it.

> **Committed chunks survive a process crash.** Each append synchronizes chunk files and atomically checkpoints the manifest before acknowledging. Recovery copies verified chunks and stays explicitly incomplete. In-flight orphan files are ignored. BLE transport and hardware power-loss qualification remain separate.

What provenance cannot prove

A hash detects changed content, not a true sensor claim. A permission string is an audit field, not a license grant. Declared alignment error must come from measured synchronization, not a convenient number.

Causal features.
A bounded baseline.

Predict whether the independently defined hold is maintained. Derive interruption probability as `1 − P(hold)`.

0.5 s

TRAILING FEATURE WINDOW

No samples after the output bin end. Warm-up windows abstain.

0.1 s

OUTPUT BIN / UPDATE STEP

This is sampling resolution, not a claim of 100 ms end-to-end response.

COMPONENT	EXACT BASELINE
Required input	EMG in microvolts, with ordered channels and matching signal-chain metadata. Every sample in the feature window must be valid.
Per-channel features	Window-centered RMS, mean absolute value, waveform length per second, RMS / calibration reference, and first-half to second-half RMS slope per second.
Calibration	One positive finite RMS reference per channel; calibration ID and source. Availability must precede the feature context. Future labelled trials cannot provide it.
Optional motion	Disabled by default. When enabled: six ordered axes; acceleration in m/s² and gyro in rad/s. Adds acceleration-vector RMS and angular-velocity RMS.
Classifier	200-tree random forest; maximum depth 8; minimum leaf size 2; balanced class weights; deterministic seed, default 42.
Comparator	Mean calibrated RMS threshold selected on training data by balanced accuracy. Reported alongside the forest on validation data.

Deliberate limits

Awake truth: separate pressure in kPa, using the selected JMS TPM-02 recording path. Protocol specifies pressure threshold, hold duration and interruption duration; short runs or missing pressure are excluded. Actual instrument calibration, synchronization and clinician-approved criteria remain unverified. See the linked reference decision.

Probabilities are uncalibrated model scores. No clinical criteria are fabricated; real training requires approved reference metadata and reproduced labels. No live cue engine is enabled. The model does not establish full exercise correctness, isolated muscle activity or tongue position.

Start with an inspectable audio baseline.

Use microphone data alone at inference. Clinical reference channels define labels; they never become hidden model inputs.

COMPONENT	IMPLEMENTED BEHAVIOR
Audio boundary	Mono, 16 kHz, normalized full-scale samples. Channel/rate/configuration mismatch abstains. Samples at or beyond declared clipping rails invalidate the window.
Context and output	2-second trailing context, producing one probability per 100 ms nonoverlapping output bin. Overlapping feature windows do not duplicate output bins.
Spectral frontend	Complete unpadded 32 ms Hann frames; 10 ms hop; 64 triangular mel filters spanning 20 Hz to 7.6 kHz at 16 kHz sampling; natural-log power floor 10^{-12}.
28 features	Log RMS; zero-crossing rate; mean and standard deviation of the first 13 orthonormal type-II DCT coefficients of log-mel energy.
Classifier	Training-only standardization followed by class-balanced logistic regression; maximum 1,000 iterations. Means, scales, coefficients and intercept are exported in JSON.

> **Silence is not an event label.** Valid unscored audio stays unlabelled. Quiet breathing, detachment, another person's sound, wake and equipment noise need independent handling and validation.

Next experiment, deliberately not implemented

The supplied brief proposes 20-second log-mel context, compact convolution layers, a bidirectional GRU and per-time-step probabilities. Bidirectional context would be permitted only for offline analysis.

Gate that experiment on permitted aligned recordings, a frozen baseline and a preregistered improvement target. Train and compare on identical participant partitions; stitch overlapping block predictions before event extraction. Add a temporal network only if it improves the relevant held-out tradeoff.

The current package does not contain a trained neural detector, pretrained apnea checkpoint or clinical AHI estimator. The 2-second baseline is a concrete first comparison, not an implementation of the proposed 20-second temporal network.

People are the split unit.
Time is the denominator.

TRAIN	Fit features' learned normalization and model weights. Both classes must have valid independent labels.
VALIDATION	Select event-probability threshold from 0.05 to 0.95 in 0.05 steps using frame balanced accuracy.
FREEZE	Save split IDs, input contract, feature order, event rules, source provenance and dataset/model digests.
TEST	Score only held-out participants with the frozen bundle. All repeated sessions from one person stay in the same partition.

Default split fractions reserve 20% each for validation and test, with at least one participant in each. Three participants only satisfy the software partition requirement; that is not a sufficient clinical study design.

EVALUATION DECISION	SEMANTICS
Unknown reference	Training/frame labels require one scored state. Mixed-boundary bins stay unlabelled. Event scoring uses original continuous intervals and separate scored coverage, preserving off-grid boundaries.
Candidate extraction	Threshold per-task event probability. Invalid or missing bins break candidates; gap merging cannot bridge unavailable data. Default minimum duration and gap merge are zero, research settings only.
Event matching	One-to-one maximum-cardinality matching, with total temporal IoU as the tie-break. Default IoU threshold: 0.3.
Censored reference	Exact reference events intersecting invalid, unscored or missing coverage are censored. Their spans and overlapping predictions are excluded consistently; counts remain visible. Unknown frame labels at an otherwise scored event boundary do not censor it.

Validation is model-selection evidence, not an unbiased performance claim. The local test command refuses changed dataset content. Independent-site evaluation needs a separately versioned external evaluation; a second random split is not an external cohort.

Implementation: evaluation.py · pipeline.py · dataset.py

Threshold and duration settings are engineering experiments, not clinical diagnostic definitions.

Abstention is an output. It is never a successful hold.

CONDITION	IMPLEMENTED RESPONSE
Missing stream / incompatible input contract	Invalid prediction with null probability and a reason. Training rejects mixed signal-chain cohorts.
Missing samples / invalid motion / declared clipping	Window abstains. Candidates cannot cross an invalid or missing output bin.
Unavailable calibration / insufficient history	Awake prediction abstains until the entire required context and usable calibration exist.
Unknown labels / excluded intervals	Inference can still run when inputs are valid. Unknown reference cannot contribute supervised training or become a negative label.
Corrupt chunks / invalid bundle / leakage	Fail explicitly. Reject checksum mismatch, malformed shapes, unsafe tree structure and shared participant IDs across partitions.
Synthetic versus real data	Training requires explicit synthetic permission; mixed manifests and synthetic/real model-replay combinations are refused. Synthetic status travels with outputs.

Required reading of every result

Reports include event sensitivity and precision, false detections per evaluable hour, usable and labelled coverage, censoring counts, duration-weighted precision-recall and Brier score, matched timing errors and temporal IoU.

Awake event metrics concern interruptions. False interruptions per minute use valid labelled event-evaluable time. Instructed-practice-only rates are not available. They are not tested coaching cues: no cue scheduler is implemented.

Confidence intervals resample participants, with 1,000 percentile-bootstrap replicates. Fewer than two participants produces no interval. Undefined denominators remain null. Insufficient evaluable data gets an explicit status; confound summaries report frame metrics and exposure.

> **Do not rename the denominator.** Candidate events per recorded or evaluable hour are not clinical AHI. Uncalibrated score, overlap or output resolution is not diagnostic certainty.

Architecture can be locked. Performance must be earned.

GATE	EVIDENCE REQUIRED TO PASS	CURRENT BOUNDARY
1 / Acquisition	Exact channel routing, ADC scaling and bandwidth; clipping/contact behavior; counter and timestamp mapping; measured drift and sustained capture with every dropout accounted for.	No hardware transport implemented or validated.
2 / Independent truth	Clinician-defined hold/interruption criterion and suitable synchronized reference. For audio: permitted recordings with scored event and exclusion intervals, original subtypes retained.	No clinical training data supplied to this build.
3 / Baseline integrity	Replay saved samples, verify units/alignment, freeze subject splits, compare simple baselines and retain realistic prevalence. Synthetic tests cannot substitute for this gate.	Software pipeline and checks implemented; real-data gate unpassed.
4 / Useful discrimination	Set minimum sensitivity, maximum false events, acceptable coverage, timing error and subgroup/confound tolerances before final testing. Choose study size and acceptable uncertainty.	Clinical owners must define these acceptance values. None invented here.
5 / Hardware transfer	Evaluate Arc recordings alongside an independent reference, across placement changes, people and recording conditions. Keep external data separate from model selection.	Public-data success alone cannot pass.
6 / Feedback release	Validated sustained-evidence threshold, confirmation interval, cooldown and measured total latency; show feedback improves practice under the approved protocol.	Live coaching disabled. No treatment recommendation path.

Security and retention

Local files only; JSON model weights avoid executable pickle. Keep identity mappings outside research sessions. Raw audio access, encryption, retention and backup rules must be supplied by the approved operating environment.

Deployment sequence

Saved-session laptop analysis first. A BLE receiver, one-phone logger, phone export parity and full-night hardware testing remain. Incremental checkpoints and process-crash recovery are implemented. No phone runtime, VM deployment or autonomous medical action is included.

Release claim today: reproducible research software with explicit failure behavior. Not a trained clinical product. No architecture can replace permitted data, measured capture quality or independent validation.